

INT396:UNSUPERVISED LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain the principles, paradigms, and real-world applications of unsupervised learning for exploratory data analysis and knowledge discovery.

CO2 :: apply clustering, similarity measures, and dimensionality reduction techniques to identify hidden structures in unlabeled datasets.

CO3 :: analyze partition-based, hierarchical, density-based, and probabilistic learning techniques to select appropriate unsupervised models for diverse problem domains.

CO4 :: evaluate clustering, association rule mining, anomaly detection, and representation learning outcomes using appropriate validation metrics and visualization techniques.

CO5 :: evaluate the effectiveness of multiple unsupervised learning techniques in solving real-world pattern discovery, customer analytics, and anomaly detection problems.

CO6 :: apply end-to-end unsupervised learning workflows involving data preprocessing, model development, visualization, interpretation, and ethical analysis for intelligent decision support.

Unit I

Foundations of Unsupervised Learning : Supervised vs. Unsupervised vs. Semi-Supervised Learning, Problem formulation for unsupervised learning, Real-life use cases: (Market segmentation, Customer behavior analysis, Anomaly & fraud detection, Pattern discovery in biological and social data). Distance & similarity metrics:(Euclidean, Manhattan, Cosine similarity, when and why distance choice matters), Supervised vs. Unsupervised vs. Semi-Supervised Learning, Problem formulation for unsupervised learning, Real-life use cases: Market segmentation, Customer behavior analysis, Anomaly & fraud detection, Pattern discovery in biological and social data, Distance & similarity metrics:Euclidean, Manhattan, Cosine similarity, when and why distance choice matters

Unit II

Partition-Based Clustering : Clustering fundamentals and assumptions, Hard vs. Soft clustering. k-Means algorithm: Objective function,Initialization strategies (Random, k-Means++), Convergence and limitations, k-Medoids (PAM) vs. k-Means, Data standardization and scaling impact, MiniBatch k-Means for large-scale datasets Cluster Validation: Inertia (WCSS), Silhouette Coefficient, Davies–Bouldin Index, Elbow method pitfalls, Clustering fundamentals and assumptions, Hard vs. Soft clustering, k-Means algorithm: Objective function,Initialization strategies (Random, k-Means++), Convergence and limitations, k-Medoids (PAM) vs. k-Means, Data standardization and scaling impact, MiniBatch k-Means for large-scale datasets, Cluster Validation: Inertia (WCSS), Silhouette Coefficient, Davies–Bouldin Index, Elbow method pitfalls

Unit III

Hierarchical & Density-Based Clustering : Hierarchical clustering: Agglomerative vs. Divisive, Linkage methods (single, complete, average, Ward), Dendrogram interpretation, Density-based clustering: DBSCAN fundamentals, e-neighborhood, MinPts, Noise and border points, Comparison: k-Means vs. Hierarchical vs. DBSCAN, Hierarchical clustering: Agglomerative vs. Divisive, Linkage methods (single, complete, average, Ward), Dendrogram interpretation, Density-based clustering: DBSCAN fundamentals, e-neighborhood, MinPts, Noise and border points, Comparison: k-Means vs. Hierarchical vs. DBSCAN

Unit IV

Dimensionality Reduction and Representation Learning : Need for dimensionality reduction, linear dimensionality reduction techniques (PCA – geometric intuition), manifold learning overview, non-linear dimensionality reduction concepts (t-SNE, UMAP – conceptual), autoencoder intuition, Need for Dimensionality Reduction, Linear Dimensionality Reduction Techniques (PCA – Geometric Intuition), Manifold Learning Overview, Non-Linear Dimensionality Reduction Concepts -t-SNE, UMAP – Conceptual, Autoencoder Intuition

Unit V

Association Rule Mining & Anomaly Detection: Association Rule Mining : Support, Confidence, Lift, Conviction, Apriori algorithm, FP-Growth algorithm, Market Basket Analysis Anomaly Detection: Anomaly vs. novelty detection, Isolation Forest, Local Outlier Factor (LOF), Applications in cybersecurity and fraud detection, Support, Confidence, Lift, Conviction, Apriori Algorithm, FP-Growth algorithm, Market Basket Analysis Anomaly Detection: Anomaly vs. novelty detection, Isolation Forest, Local Outlier Factor (LOF), Applications in Cybersecurity and Fraud Detection

Unit VI

Evaluation and Applications of Unsupervised Learning : Internal and external clustering validation metrics, silhouette score and cohesion–separation intuition, stability-based evaluation, interpretability challenges in unsupervised learning, real-world case studies, ethical considerations in pattern discovery, Internal and External Clustering Validation Metrics, Silhouette Score and Cohesion–Separation Intuition, Stability-Based Evaluation, Interpretability Challenges in Unsupervised Learning, Real-World Case Studies, Ethical Considerations in Pattern Discovery, Case Study : Visualizing handwritten digits (MNIST) or customer segmentation data

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Project Component : Create a visualization dashboard comparing PCA vs. t-SNE results

List of Practicals / Experiments:

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- Analyze customer behavior patterns by performing exploratory data analysis and evaluating Euclidean, Manhattan, and Cosine similarity measures on an unlabeled customer dataset.
- Develop a customer segmentation solution by implementing and comparing k-Means and k-Medoids clustering techniques on the Mall Customers dataset.
- Design a scalable clustering model by implementing k-Means from scratch and validating its performance using the Scikit-learn framework.
- Investigate the influence of data preprocessing by examining the effect of feature scaling and standardization on clustering quality and model convergence.
- Determine the optimal clustering configuration by applying Elbow Method, Silhouette Score, and Davies–Bouldin Index on a real-world marketing dataset.
- Analyze hierarchical relationships in unlabelled data by constructing dendrograms and comparing multiple linkage strategies using Hierarchical Clustering.
- Develop a density-based clustering solution by applying DBSCAN and evaluating the impact of ϵ (epsilon) and MinPts parameters on cluster formation and noise detection.
- Evaluate clustering techniques for diverse data distributions by benchmarking k-Means, Hierarchical Clustering, and DBSCAN using clustering validation metrics.
- Develop an efficient feature representation model by applying Principal Component Analysis (PCA) to reduce dimensionality while preserving maximum variance.
- Compare dimensionality reduction techniques by visualizing high-dimensional datasets using PCA, t-SNE, and UMAP for exploratory data analysis.
- Extract actionable business insights by implementing Apriori and FP-Growth algorithms for Market Basket Analysis and interpreting association rules.
- Develop an anomaly detection framework by comparing Isolation Forest and Local Outlier Factor (LOF) for identifying unusual observations in cybersecurity or financial datasets.
- Assess the quality of unsupervised learning models using clustering validation metrics, visualization techniques, and stability analysis to support model selection.
- Develop an end-to-end unsupervised learning solution by integrating data preprocessing, clustering, dimensionality reduction, anomaly detection, visualization, evaluation, and presentation of insights for a real-world application in healthcare, finance, education, agriculture, cybersecurity, or retail.

Text Books:

1. HANDS-ON UNSUPERVISED LEARNING USING PYTHON: HOW TO BUILD APPLIED MACHINE LEARNING SOLUTIONS FROM UNLABELED DATA by ANKUR A. PATEL, O'REILLY

References:

1. MODELS AND ALGORITHMS FOR UNSUPERVISED LEARNING: PRACTICAL UNSUPERVISED MACHINE LEARNING by VAIBHAV VERDHAN, Manning Publication
2. KICKSTART UNSUPERVISED MACHINE LEARNING: MASTER UNSUPERVISED MACHINE LEARNING THROUGH PATTERN DISCOVERY, CLUSTERING, AND DIMENSIONALITY REDUCTION TO BUILD INTELLIGENT, REAL-WORLD APPLICATIONS by DR. NIMRITA KOUL, AVA PUBLISHING

